Unit: 6 Unit 8: Quadratics
Topic: Definition of Quadratics and Standard Form

Name: ________________
Date: __________

Quiz (Carolina Reaper)

Q1. What is the standard form of a quadratic function?
(A) $ax^2 + bx + c$
(B) $ax + bx^2 + c$
(C) $ax^2 - bx + c$
(D) $ax + b$
(E) $ax^2 + c$

Q2. Which of the following is a quadratic function in standard form?
(A) $x^2 + 2x - 3$
(B) $2x + 1$
(C) $x^3 - 2x^2 + x - 1$
(D) $x^2 - 4$
(E) $2x^2 + 3x - 1$

Q3. What is the coefficient of the x^2 term in the quadratic function $3x^2 + 2x - 1$?
(A) 1
(B) 2
(C) 3
(D) -1
(E) -2

Q4. Which of the following quadratic functions has a negative leading coefficient?
(A) $x^2 + 2x + 1$
(B) $-x^2 + 2x - 1$
(C) $x^2 - 2x + 1$
(D) $2x^2 + 2x + 1$
(E) $-2x^2 + 2x - 1$

Q5. What is the value of a in the quadratic function ax^2+2x-3 if the function has a maximum value?
(A) 1
(B) -1
(C) 2
(D) -2
(E) 0

Q6. Which of the following is a key feature of the quadratic function $x^2 + 2x + 1$?
(A) Minimum value at $(0, 1)$
(B) Maximum value at $(0, 1)$
(C) Minimum value at $(-1, 0)$
(D) Maximum value at $(-1, 0)$
(E) No maximum or minimum value

Q7. What is the axis of symmetry of the quadratic function $x^2 + 2x + 1$?
(A) $x = -1$
(B) $x = 1$
(C) $x = 0$
(D) $x = -2$
(E) $x = 2$

Q8. Which of the following quadratic functions has its vertex at $(0, 0)$?
(A) $x^2 + 2x + 1$
(B) $x^2 - 2x + 1$
(C) x^2
(D) $x^2 + 1$
(E) $x^2 - 1$

Open Ended Questions (FRQ)

Q9. Describe the key features of the quadratic function $-x^2 + 2x + 1$. Be sure to include the vertex, axis of symmetry, and any maximum or minimum values.

Q10. Graph the quadratic function $x^2 + 2x + 1$ and identify its key features. Be sure to label the vertex, axis of symmetry, and any maximum or minimum values.

Chef's Tips

- Remind students to show all work and explain their reasoning.
- Encourage students to use graphing calculators to visualize the quadratic functions.
- Emphasize the importance of identifying key features, such as vertex and axis of symmetry.